

8.2.7

C
D₁ 与 滤波电容^V极性都反了, 都应该对调一下

8.2.9

$$U_{Z1} = 1.75V \quad \text{设 } R_i' = R_1 + R_{pL}$$

$$U_0 = \frac{U_{Z1}}{R_i'} (R_1 + R_2 + R_p)$$

$$\textcircled{1} R_i' = R_1 \text{ 时 } U_0 = U_{0\max} = 2.94V$$

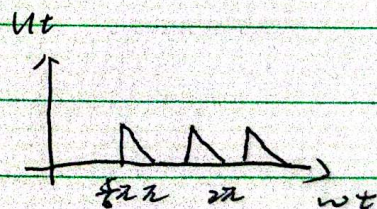
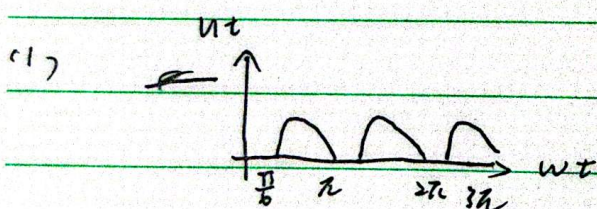
$$\textcircled{2} R_i' = R_1 + R_{p\max} \text{ 时 } U_0 = U_{0\min} = 1.399V$$

8.3.2

$$U_L = 0.9 U_2 \frac{1 + \cos \alpha}{2} \quad \frac{1 + \cos \alpha}{2} = \frac{1}{2} \frac{1 + \cos 90^\circ}{2} \quad \alpha = 90^\circ$$

$$U_2 = 50 / (0.9 \frac{1 + \cos 90^\circ}{2}) = 55.6V$$

8.3.3



$$(2) \quad U_L = 0.9 U_2 \frac{1 + \cos \alpha}{2}$$

$$U_L / \alpha = 30^\circ = 41.99V \quad U_L / \alpha = 150^\circ = 3.01V$$

$\therefore$ 范围由 3.01 ~ 41.99 V.

$$(3) \quad U_{Tm} = 2\sqrt{2} U_2 = 2 \times \sqrt{2} \times 50 = 141.4V$$

$$I_{Tm} = \frac{1}{2} I_L = \frac{1}{2} \frac{U_L}{R_L} = 1.05A$$

9.1.1

$$P_{cu} = I^2 R = 0.5^2 \times 60 = \cancel{15W} 15W$$

$$P_{Fe} = P - P_{cu} = 25W$$

9.2.1

$$k = \frac{U_{IN}}{U_{2N}} = \frac{220}{36} = 6.11$$

$$\therefore I_{2N} = k I_{1N} = \cancel{6.11} \times 6.11 \times 9.1 = 55.6A$$